

1. Concept of Percentage:- By a certain percent, we mean that many hundredths. Thus, x percent means x hundredths, written as $x\%$.

To express $x\%$ as a fraction :

$$\text{We have } x\% = \frac{x}{100},$$

$$\text{Thus, } 20\% = \frac{20}{100} = \frac{1}{5} \text{ etc.}$$

To express $\frac{a}{b}$ as a percent :

$$\text{We have } \frac{a}{b} \times 100 = \left(\frac{a}{b} \times 100 \right) \%$$

$$\text{Thus } \frac{1}{4} = \left(\frac{1}{4} \times 100 \right) \% = 25\%.$$

2. If the price of commodity increases by $R\%$, then the reduction in consumption so as to not to increase the expenditure is

$$\left[\frac{R}{(100 + R)} \times 100 \right] \%$$

If the price of ~~the~~ a commodity decreases by $R\%$, then the increase in consumption so as to decrease the expenditure is

$$\left[\frac{R}{(100 - R)} \times 100 \right] \%$$

3. Results on Population :- Let the population of a town be P now and suppose it increases at the rate of $R\%$ per annum, then:

1. Population after n years $= P \left(1 + \frac{R}{100} \right)^n$

2. Population n years ago $= \frac{P}{\left(1 + \frac{R}{100} \right)^n}$

4. Results on Depreciation :- Let the present value of a machine be P . Suppose it depreciates at the rate of $R\%$ per annum. Then:

1. Value of the machine after n years $= P \left(1 - \frac{R}{100} \right)^n$

2. Value of the machine n years ago $= \frac{P}{\left(1 - \frac{R}{100} \right)^n}$

5. If A is $R\%$ more than B , then B is less than A by

$$\left[\frac{R}{(100 + R)} \times 100 \right] \%$$

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If A is R% less than B, then B is more than A by

$$\left[\frac{R}{(100-R)} \times 100 \right] \%$$

Ex:-1 Express 56% as a fraction.

Solⁿ $56\% = \frac{56}{100} = \frac{14}{25}$

Ex:-2 Express 0.2% as a decimal.

Solⁿ $0.2\% = \frac{0.2}{100} = 0.002$

Ex:-3 Express $6\frac{3}{4}$ as rate percent.

Solⁿ $6\frac{3}{4} = \frac{27}{4} = \left(\frac{27}{4} \times 100 \right) \% = 675\%$

Ex:-4 Evaluate 28% of 450 + 45% of 280

Solⁿ $\left(\frac{28}{100} \times 450 + \frac{45}{100} \times 280 \right) = (126 + 126) = 252$

Ex:-5 2 is what percent of 50?

Solⁿ Required percentage = $\left(\frac{2}{50} \times 100 \right) \% = 4\%$

Ex:-6 Find the missing figures:-

(i) ?% of 25 = 2.125

(ii) 0.25% of ? = 0.04

Solⁿ (i) Let x% of 25 = 2.125

$$\Rightarrow \frac{x}{100} \times 25 = 2.125 \Rightarrow x = (2.125 \times 4) = 8.5$$

(ii) Let 0.25% of x = 0.04

$$\Rightarrow \frac{0.25}{100} \times x = 0.04$$

$$\Rightarrow x = \frac{0.04 \times 100}{0.25} = 16.$$

Ex:-7 Which is greatest in $16\frac{2}{3}\%$, $\frac{2}{15}$ and 0.17?

Solⁿ $16\frac{2}{3}\% = \left(\frac{50}{3} \times \frac{1}{100}\right) = \frac{1}{6} = 0.166$

$$\Delta \frac{2}{15} = 0.133$$

Clearly 0.17 is the greatest.

Ex:-8 If the sales tax be reduced from $3\frac{1}{2}\%$ to $3\frac{1}{3}\%$, then what difference does it make to a person who purchases an article with marked price of Rs. 8400?

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Soln Required difference = $\left(\frac{31}{2}\% \text{ of Rs } 8400\right) - \left(\frac{31}{3}\% \text{ of Rs } 8400\right)$

$$= \left(\frac{7}{2} - \frac{10}{3}\right)\% \text{ of Rs } 8400 = \frac{1}{6}\% \text{ of Rs } 8400$$

$$= \text{Rs.} \left(\frac{1}{6} \times \frac{1}{100} \times 8400\right) = \text{Rs. } 14.$$

Ex:-9 An inspector rejects 0.08% of the meters as defective. How many will be examine to reject 2?

Soln Let the number of meters to be examined be x .

Then $0.08\% \text{ of } x = 2 \Rightarrow \left(\frac{8}{100} \times \frac{1}{100} \times x\right) = 2$

$$\Rightarrow x = \frac{2 \times 100 \times 100}{8} = 2500.$$

Ex:-11 Difference of two numbers is 1660. If 7.5% of one number is 12.5% of the other number, find the two numbers.

Soln Let the numbers be x and y . Then,

$$7.5\% \text{ of } x = 12.5\% \text{ of } y$$

$$\Rightarrow x = \frac{125}{75} y = \frac{5}{3} y.$$

$$\text{Now, } x - y = 1660 \Rightarrow \frac{5}{3} y - y = 1660 \Rightarrow \frac{2}{3} y = 1660$$

$$\Rightarrow y = 1660 \times \frac{3}{2} = 2490.$$

$\therefore$ One number = 2490, second number = $\frac{5}{3} y = 4150$

Ex-12 In expressing a length 81.472 km as nearly as possible with three significant digits, find the percentage error:

Solⁿ Error = $(81.5 - 81.472) \text{ km} = \cancel{0.028} 0.028.$

$\therefore$ Required percentage = $\left(\frac{0.028}{81.472} \times 100 \right) \% = 0.03\%$

Ex-13 In an election between two candidates, 75% of the voters cast their votes, out of which 2% of the votes were declared invalid. A candidate got 9261 votes which were 75% of the total valid votes. Find the total number of votes enrolled in that election.

Solⁿ Let the total number of votes enrolled be x .

Number of votes cast = 75% of x .

Valid votes = 98% of (75% of x)

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$$\therefore 75\% \text{ of } [98\% \text{ of } (75\% \text{ of } x)] = 9261$$

$$\Rightarrow \left(\frac{75}{100} \times \frac{98}{100} \times \frac{75}{100} \times x \right) = 9261$$

$$\Rightarrow x = \frac{9261 \times 100 \times 100 \times 100}{75 \times 98 \times 75} = 16800$$

3087 10 29 4 3 4
3, 4, 8,

Ex:-14 Shobha's Mathematics Test had 75 problems i.e. 10 arithmetic, 30 algebra and 35 geometry problems. Although she answered 70% of the arithmetic, 40% of the algebra and 60% of the geometry problems correctly, she did not pass the test because she got less than 60% of the problems right. How many more questions she would have needed to answer correctly to earn a 60% passing grade?

Solⁿ Number of questions attempted correctly =

$$(70\% \text{ of } 10 + 40\% \text{ of } 30 + 60\% \text{ of } 35)$$

$$= (7 + 12 + 21) = 40.$$

Questions to be answered correctly for 60% grade =
 $60\% \text{ of } 75 = 45$.

Required number of ~~two~~ questions = $45 - 40 = 5$.

Ex-15 If 50% of $(x-y) = 30\%$ of $(x+y)$, then what percent of x is y ?

Soln 50% of $(x-y) = 30\%$ of $(x+y)$

$$\Rightarrow \frac{50}{100} (x-y) = \frac{30}{100} (x+y)$$

$$\Rightarrow 5(x-y) = 3(x+y) \Rightarrow 2x = 8y \Rightarrow x = 4y.$$

$$\therefore \text{Required percentage} = \left(\frac{y}{x} \times 100 \right) \% = \left(\frac{y}{4y} \times 100 \right) \% = 25\%$$

Ex-16 Mr. Jones gave 40% of the money he had, to his wife. He also gave 20% of the remaining amount to each of his three sons. Half of the amount now left was ~~spent~~ spent on miscellaneous items and the remaining amount of Rs. 12,000 was deposited in the bank. How much ~~money~~ money did Mr. Jones have initially?

Solⁿ

Let the initial amount with Mr. Jones be Rs. x . Then,

$$\text{Money given to wife} = \text{Rs. } \frac{40}{100} x = \text{Rs. } \frac{2x}{5}$$

$$\text{Balance} = \text{Rs. } \left(x - \frac{2x}{5} \right) = \text{Rs. } \frac{3x}{5}$$

$$\begin{aligned} \text{Money given to 3 sons} &= \text{Rs. } \left[3x \left(\frac{20}{100} \times \frac{3x}{5} \right) \right] \\ &= \text{Rs. } \frac{9x}{25} \end{aligned}$$

$$\text{Balance} = \text{Rs. } \left(\frac{3x}{5} - \frac{9x}{25} \right) = \text{Rs. } \frac{6x}{25}$$

$$\text{Amount deposited in bank} = \text{Rs. } \left(\frac{1}{2} \times \frac{6x}{25} \right) = \text{Rs. } \frac{3x}{25}$$

$$\therefore \frac{3x}{25} = 12000 \Rightarrow x = \left(\frac{12000 \times 25}{3} \right) = 100000$$

So, Mr. Jones initially had Rs. 1,00,000 with him.

Exi-17 A salesman's commission is 5% on all sales up to Rs. 10,000 and 4% on all sales

Solⁿ Let his total sale be Rs. x .

Now,

$$(\text{Total Sales}) - (\text{Commission}) = \text{Rs. } 31,100$$

$$\therefore x - [5\% \text{ of } 10,000 + 4\% \text{ of } (x - 10,000)] = 31,100.$$

$$\Rightarrow x - \left[\frac{5}{100} \times 10,000 + \frac{4}{100} [x - 10,000] \right] = 31,100.$$

$$\Rightarrow x - 500 - \frac{(x - 10,000)}{25} = 31,100$$

$$\Rightarrow x - \frac{x}{25} = 31,600 \Rightarrow x = 32500.$$

Ex:-18 'Raman's salary was decreased by 50% and subsequently increased by 50%. How much percent does he lose?

Solⁿ Let original salary = Rs. 100.

$$\text{New final salary} = 150\% \text{ of } (50\% \text{ of Rs. } 100)$$

$$= \text{Rs. } \left(\frac{150}{100} \times \frac{50}{100} \times 100 \right) = \text{Rs. } 75.$$